

Chemotherapy for acute myeloid leukaemia (AML)

Chemotherapy is the main treatment for AML. It uses anti-cancer (cytotoxic) drugs to destroy or damage leukaemia cells.

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What is chemotherapy?

Chemotherapy uses anti-cancer (cytotoxic) drugs to destroy or damage leukaemia cells. These drugs affect with the way [leukaemia cells](#) grow and divide.

About chemotherapy for acute myeloid leukaemia (AML)

Chemotherapy is the main treatment for [acute myeloid leukaemia](#) (also called AML or AML leukemia). You usually have it in 2 main phases. You may also have a [targeted therapy](#) drug along with chemotherapy.

If you are not well enough to cope with intensive chemotherapy, your doctor may advise less intensive chemotherapy.

If you have [acute promyelocytic leukaemia \(APL\)](#), your treatment will include drugs called ATRA or ATO.

Induction treatment

The aim of induction treatment is to quickly get rid of the leukaemia cells. This allows your [bone marrow](#) to work normally again. This is called remission. You usually need to stay in hospital during treatment and possibly for a few weeks after until your blood cells recover.

You will need a lot of support from nursing and medical staff. They will monitor you closely for side effects, such as infection. You may need blood or platelet transfusions because your blood cell count will be low for a few weeks. You may also need antibiotics and antiviral drugs to prevent or treat infection. This is called [supportive therapy](#).

You may need 1 or 2 cycles of induction treatment. A cycle of treatment usually lasts about 10 days. You may be able to go home between treatments. You usually have a combination of 2 or 3 drugs. Induction chemotherapy usually includes:

- [cytarabine](#)
- [daunorubicin](#).

Sometimes you have cytarabine along with another chemotherapy drug called [fludarabine](#) (Fludara[®]).

Depending on the results of tests on the leukaemia cells, you may also have targeted therapy drugs.

You may have a chemotherapy drug called liposomal cytarabine–daunorubicin (Vyxeos[®]) if:

- AML is linked to previous cancer treatment or to [myelodysplastic syndrome \(MDS\)](#)
- the leukaemia cells have certain genetic changes.

Testing for remission

After chemotherapy, your marrow and blood cells start to recover. You will have a [bone marrow biopsy](#) and blood tests to check whether you are in remission. If tests show very small numbers of leukaemia cells (minimal residual disease) or none, your doctor will say you are in remission.

Consolidation treatment

You usually have consolidation treatment after 2 cycles of induction, if you are in remission. It aims to get rid of any remaining leukaemia cells and prevent the leukaemia coming back. It starts when your blood cell count is normal. You may have consolidation treatment as an outpatient. You have it for a few months.

The most common drug used in this phase is high-dose cytarabine (HiDAC). You usually have it on its own or sometimes with other chemotherapy drugs.

Some people go on to have a stem cell transplant after 1 or 2 cycles of chemotherapy. This is usually a [donor \(allogeneic\) transplant](#), but it may be from [your own cells \(autologous\)](#). Your doctor will tell you more about whether this is suitable for you.

Less intensive chemotherapy

You will have lower doses of chemotherapy or other drugs. The aim is to control AML for as long as possible and to avoid the side effects of intensive chemotherapy. If you have other medical conditions or health problems, this type of treatment may be an option for you. It has a lower risk of serious side effects and may be easier to cope with. You may be able to have some treatments as an outpatient.

Some drugs that may be used include:

- low-dose cytarabine
- [azacitidine](#).

Some people have a targeted therapy drug called [venetoclax](#) (Venclyxto®) with chemotherapy.

Your doctors will take regular blood and bone marrow samples to see how treatment is working. This helps them decide on any further treatment you may need.

Side effects of chemotherapy

Chemotherapy can cause side effects. Your doctor, nurse or pharmacist will explain what to expect. This depends on the intensity of your treatment and the drugs you have. Different drugs cause different side effects. They will explain how side effects can be controlled or managed. They will also talk to you about the risk of possible [late effects](#).

Your doctors and nurses will monitor you carefully. It is important Always to tell them about any side effects you have.

We have more information about the [side effects of chemotherapy](#).

About our information

References



Below is a sample of the sources used in our acute myeloid leukaemia (AML) information. If you would like more information about the sources we use, please contact us at cancerinformationteam@macmillan.org.uk

Heuser M et al. Acute myeloid leukaemia in adult patients: ESMO Clinical Practice Guidelines for diagnosis, treatment and follow-up. Annals of Oncology, 2020, vol 3, issue 6. Available from www.esmo.org/ [accessed 2021].

NICE (National Institute for Health and Care Excellence). Blood and bone marrow cancers. Available from <https://pathways.nice.org.uk/pathways/blood-and-bone-marrow-cancers> [accessed August 2021].

Reviewers



This information has been written, revised and edited by Macmillan Cancer Support's Cancer Information Development team. It has been reviewed by expert medical and health professionals and people living with cancer. It has been approved by Senior Medical Editor, [Dr Anne Parker](#), Consultant Haematologist.

Our cancer information has been awarded the [PIF TICK](#). Created by the Patient Information Forum, this quality mark shows we meet PIF's 10 criteria for trustworthy health information.

The language we use

We want everyone affected by cancer to feel our information is written for them.

We want our information to be as clear as possible. To do this, we try to:

- use plain English
- explain medical words
- use short sentences
- use illustrations to explain text
- structure the information clearly
- make sure important points are clear.

We use gender-inclusive language and talk to our readers as 'you' so that everyone feels included. Where clinically necessary we use the terms 'men' and 'women' or 'male' and 'female'. For example, we do so when talking about parts of the body or mentioning statistics or research about who is affected.

You can read more about how we produce our information [here](#).

Date reviewed

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ⓘ This content is currently being reviewed. New information will be coming soon.

Our cancer information meets the [PIF TICK](#) quality mark.

This means it is easy to use, up-to-date and based on the latest evidence. Learn more about [how we produce our information](#).

How we can help

Macmillan Support Line

The Macmillan Support Line is a free and confidential service for people living with and affected by cancer. If you need to talk, we'll listen.



Call us free on 0808 808 0000

8am to 8pm, 7 days a week. You can talk to our cancer specialists about anything.



Email the Macmillan Support Line

Email the Macmillan Support Team if you have any questions.

Chat online

Available 8am to 7.45pm, 7 days a week. For confidential, real-time support if you don't want to call.



Online Community

Chat online anonymously to others who understand what you are going through. Our community is available 24/7 and has dedicated forums where you can get advice and ask our experts.



Find cancer support services in your area

What's going on near you? Find out about support groups, where to get information and how to get involved with Macmillan where you live.

Get help